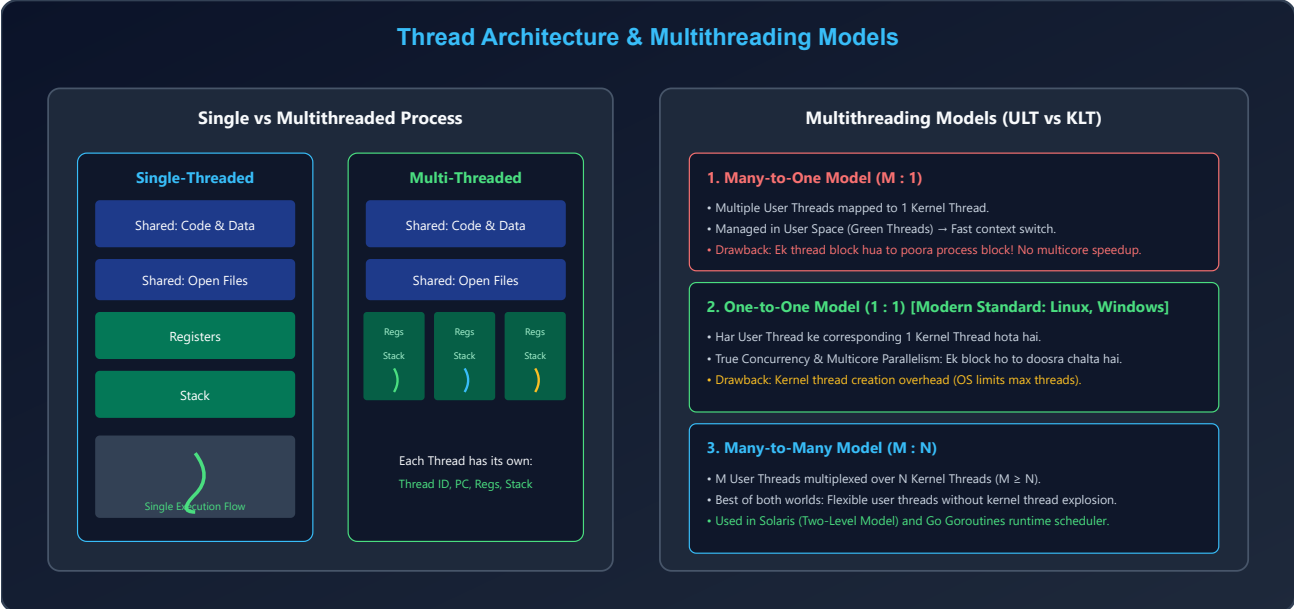


Module 03: Threads & Multithreading Models

1. Single vs Multithreaded Process Architecture



2. ULT vs KLT Comparative Table

Feature	User-Level Threads (ULT)	Kernel-Level Threads (KLT)
Managed by	User-level thread library	Operating System Kernel
Context Switch	Fast (No kernel trap)	Slower (Privilege switch Ring 3 → 0)
System Call Blocking	Entire process is blocked	Only calling thread is blocked
Multicore Parallelism	No (Cannot utilize multiple CPUs)	Yes (Independent core scheduling)

3. Multithreading Models

- Many-to-One (M:1):** Fast, but blocking one blocks all.
- One-to-One (1:1):** High concurrency, modern OS standard (Linux, Windows).
- Many-to-Many (M:N):** Hybrid model, optimal thread multiplexing (Go Goroutines).